

WEEK 2

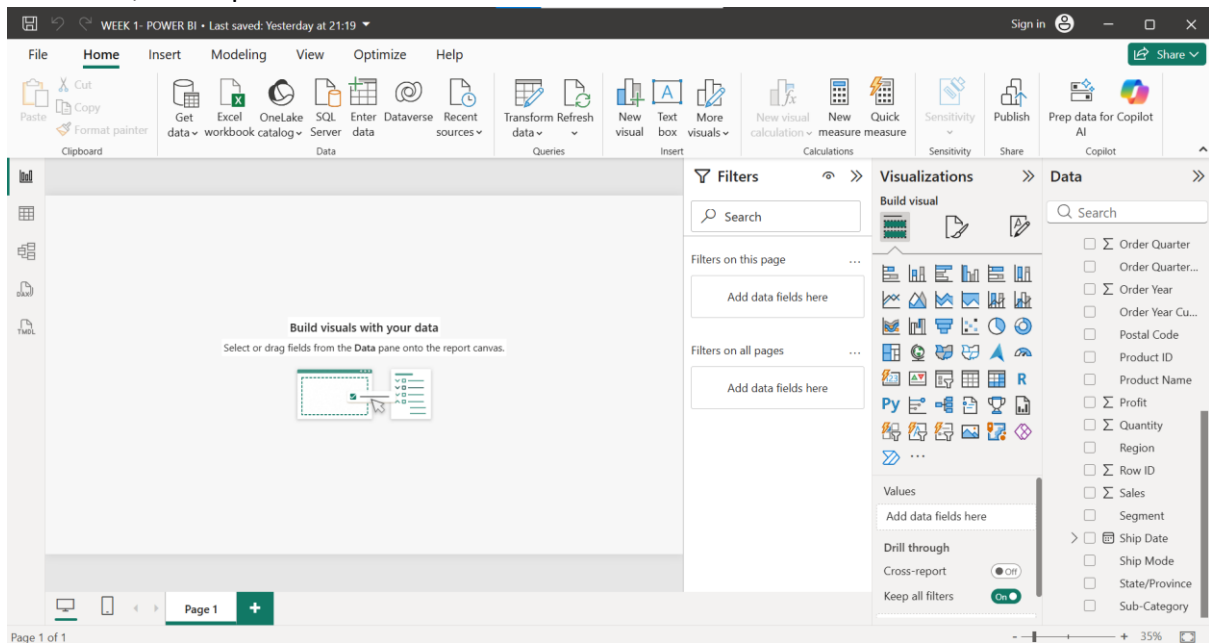
DATA MODELING, DAX AND SALES DASHBOARD

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1. Power BI Desktop Workspace Setup

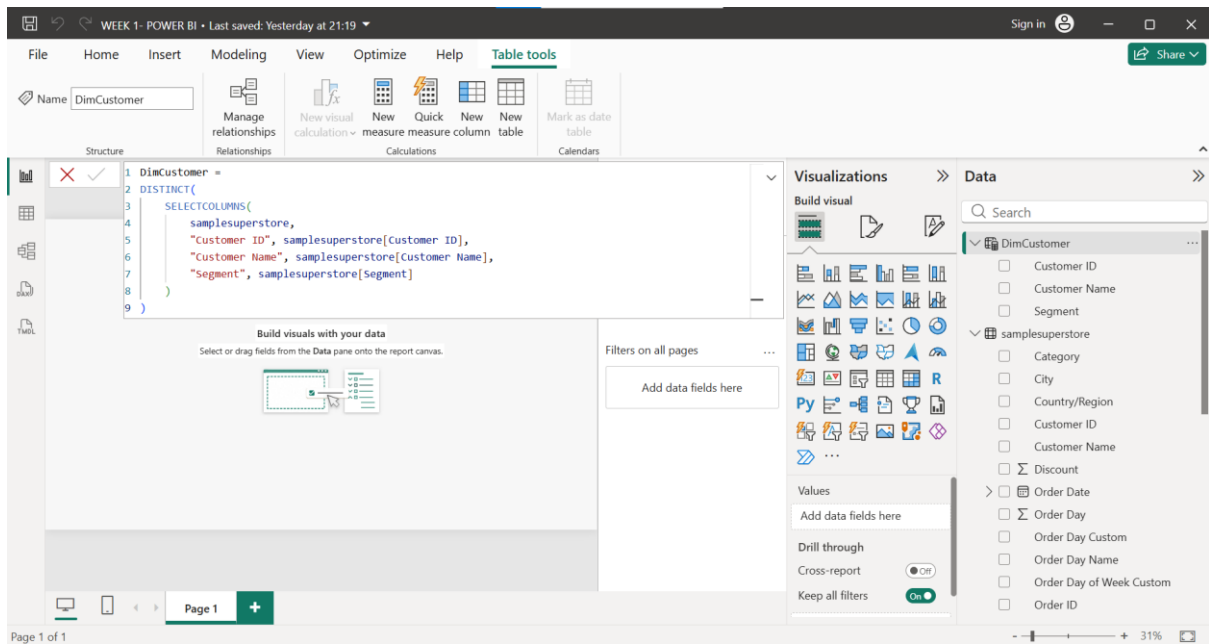
This screenshot shows the **Power BI Desktop workspace** prepared for building the Week 1 sales analysis report. The data fields from the Superstore dataset are available in the Data pane, while the Visualizations and Filters panels are ready for creating interactive charts, measures, and report elements.



Power BI Desktop workspace with the Superstore dataset loaded and ready for report creation.

2. DimCustomer Dimension Table Creation

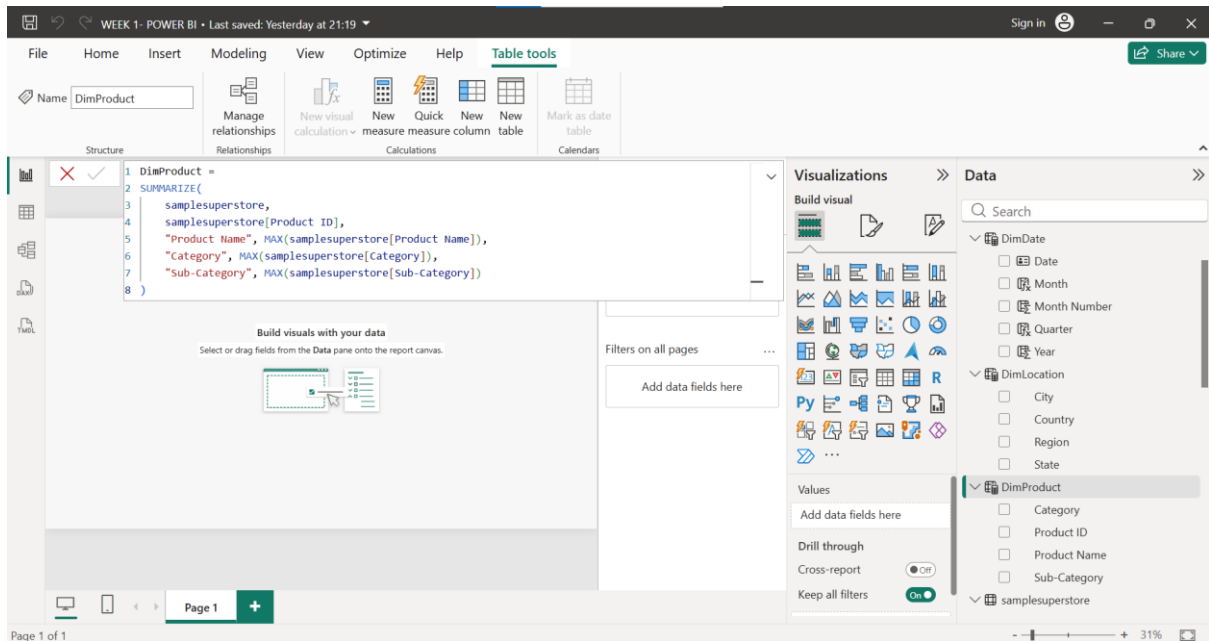
This screenshot shows the creation of a **DimCustomer dimension table** in Power BI using DAX. The DISTINCT and SELECTCOLUMNS functions are used to extract unique Customer ID, Customer Name, and Segment values from the original sales table, supporting a structured data model.



Created the DimCustomer table using DAX to organize unique customer information.

3. DimProduct Dimension Table Creation

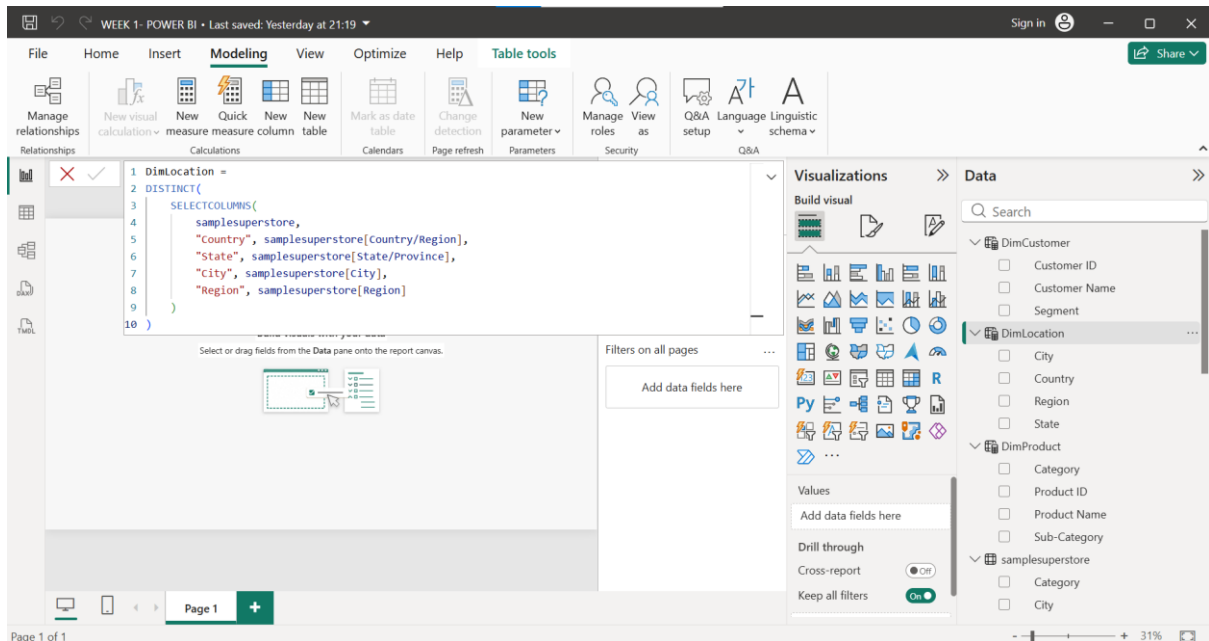
This screenshot shows the creation of the **DimProduct** table in Power BI using DAX. The table organizes product information such as **Product ID**, **Product Name**, **Category**, and **Sub-Category** from the original sales dataset, helping to build a structured data model.



Created the DimProduct table using DAX to organize product and category information.

4. DimLocation Dimension Table Creation

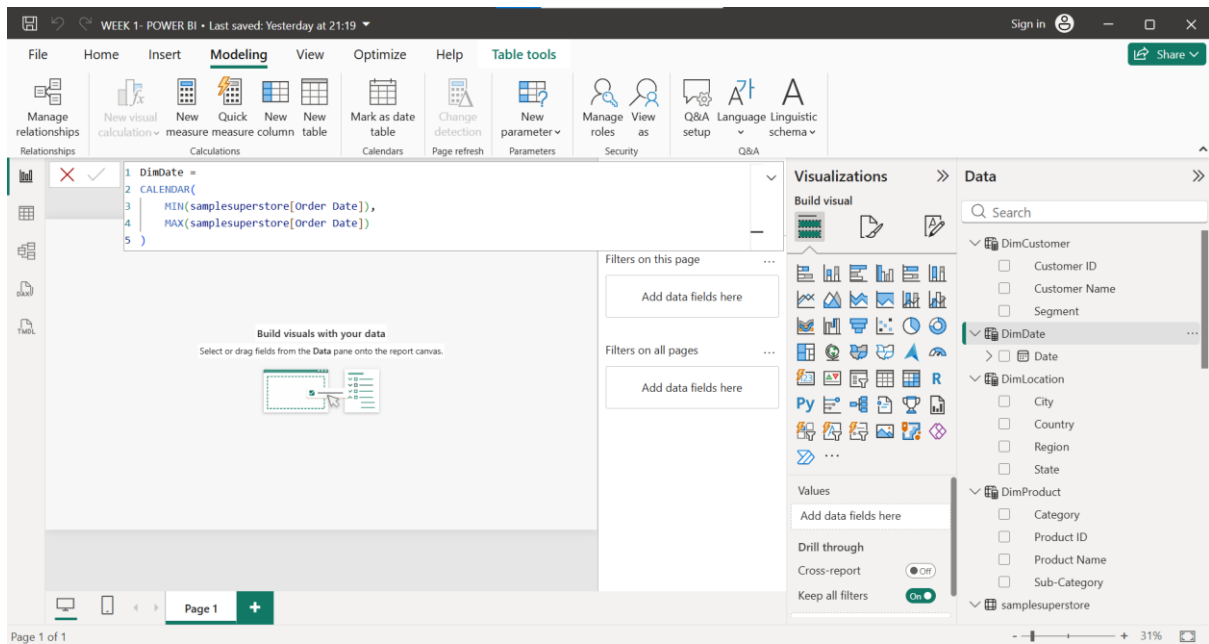
This screenshot shows the creation of the **DimLocation** table in Power BI using DAX. The table organizes geographical information such as **Country, State, City, and Region** from the Superstore dataset, helping separate location-related details into a dedicated dimension table.



Created the DimLocation table using DAX to organize geographical and regional information.

5. DimDate Table Creation

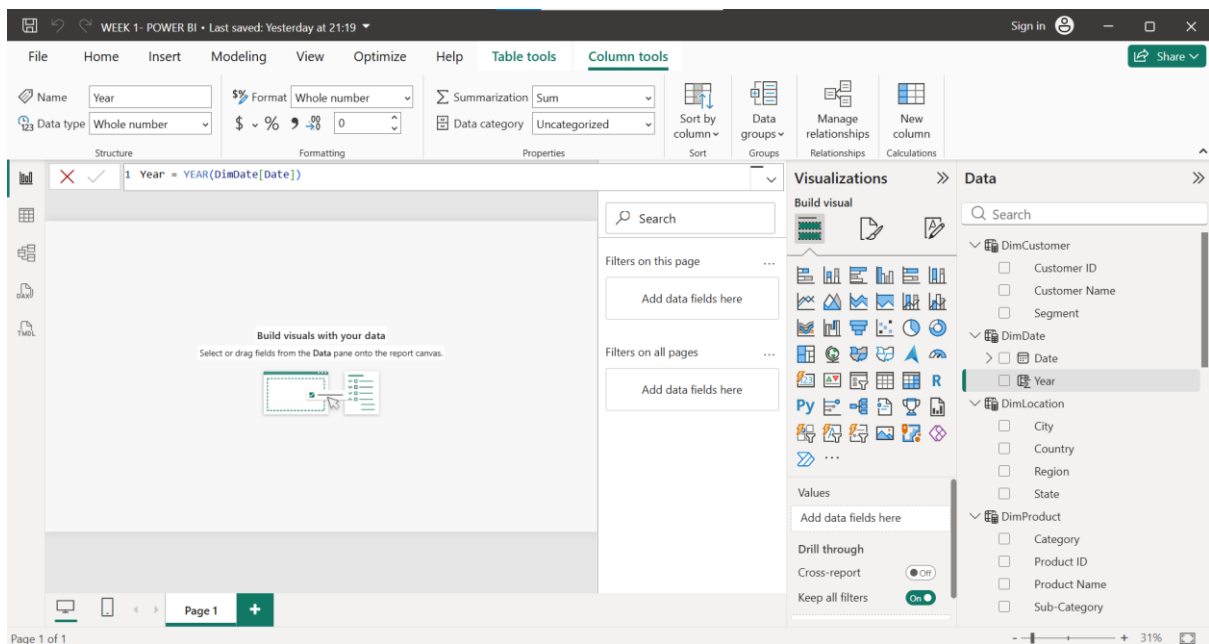
Created the **DimDate** table using DAX with a calendar range based on the minimum and maximum Order Date values from the Superstore dataset.



Created the DimDate table to support time-based analysis in Power BI.

6. Year Column Creation in DimDate

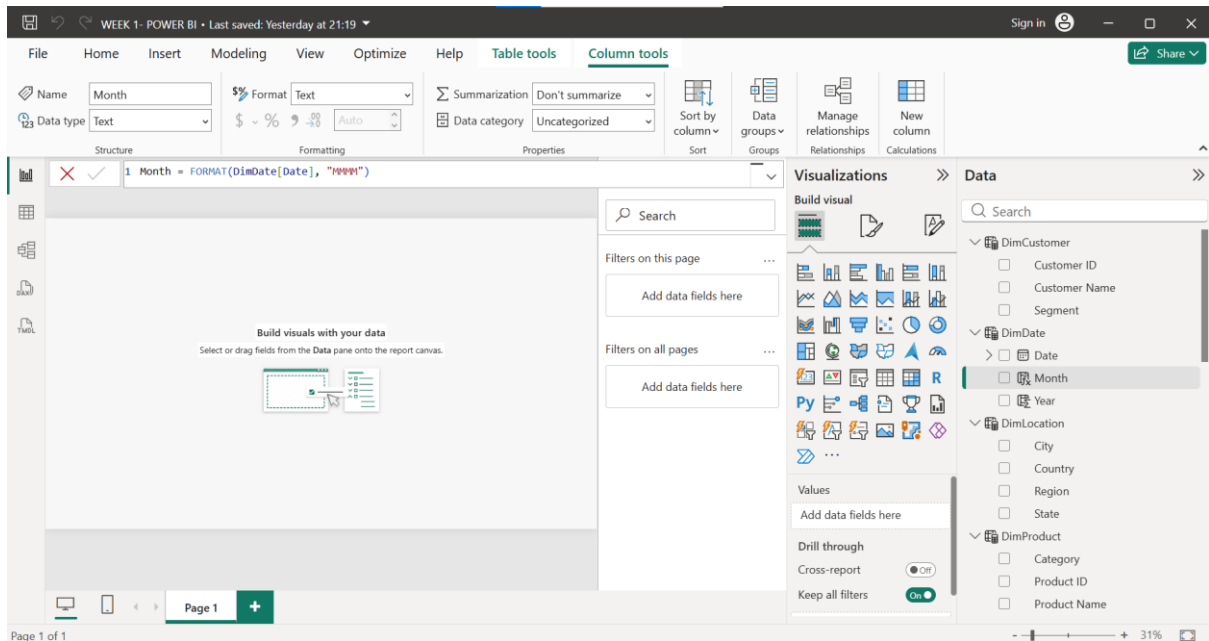
Created a **Year column** in the DimDate table using DAX to support time-based analysis and filtering in the Power BI report.



Created a Year column in the DimDate table for time-based analysis.

7. Month Column Creation in DimDate

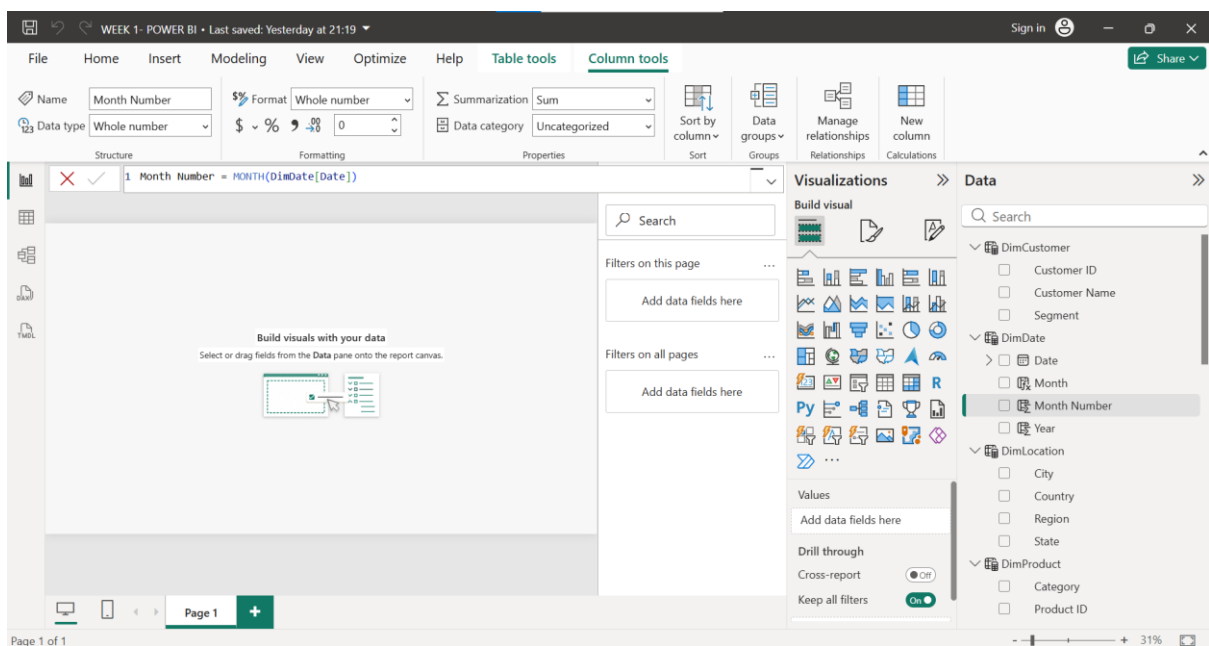
Created a **Month column** in the DimDate table using DAX to extract the month name from the Date column. This helps in analyzing and displaying sales data on a monthly basis.



Created a Month column in DimDate for monthly time-based analysis.

8. Month Number Column Creation in DimDate

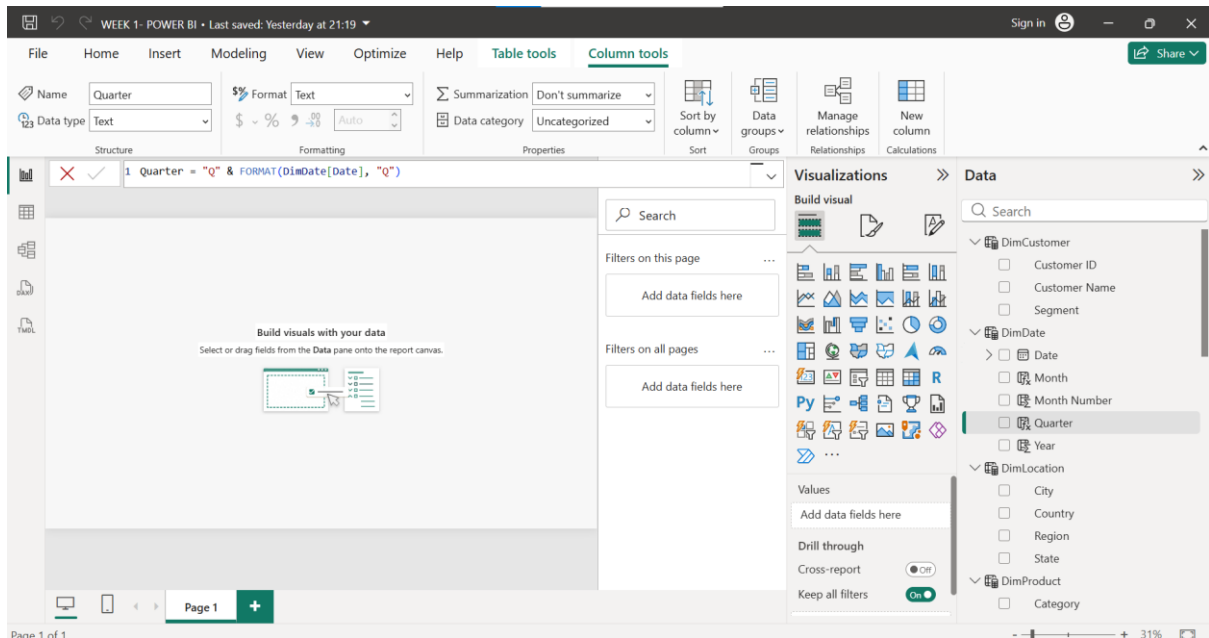
Created a **Month Number column** in the DimDate table using DAX to assign a numeric value from **1 to 12** for each month. This helps arrange months in the correct chronological order.



Created a Month Number column for proper chronological sorting of months.

9. Quarter Column Creation

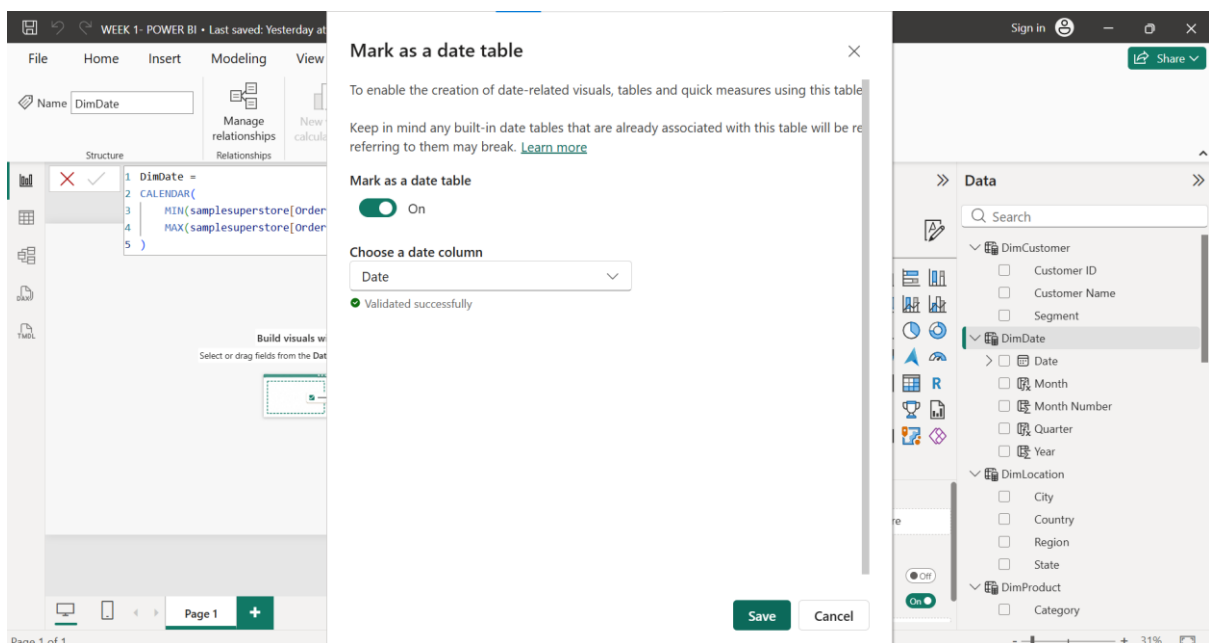
Created a **Quarter column** in the DimDate table using DAX to categorize dates into **Q1, Q2, Q3, and Q4**. This helps analyze sales and business performance across different quarters.



Created a Quarter column in DimDate for quarterly time-based analysis.

10 . Mark Date Table

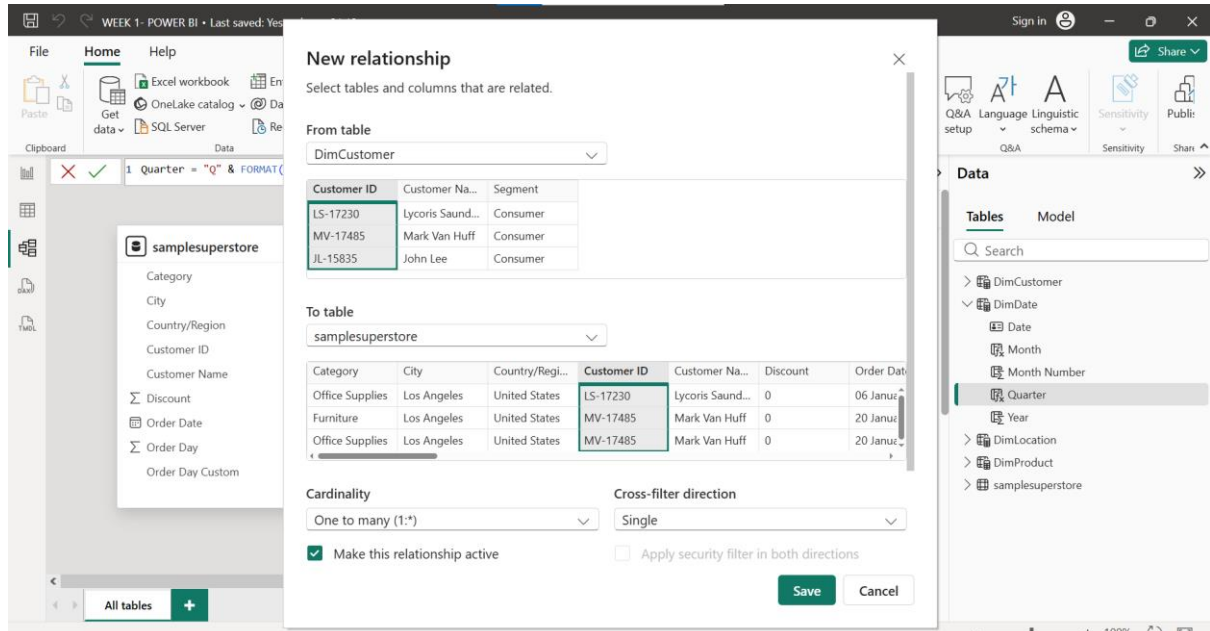
Marked the **Date column** in the DimDate table as the official **Date column** in Power BI. This enables proper date-based calculations, filtering, and time-intelligence analysis.



Marked the Date column as the official date field for time-based analysis.

11. Customer Relationship

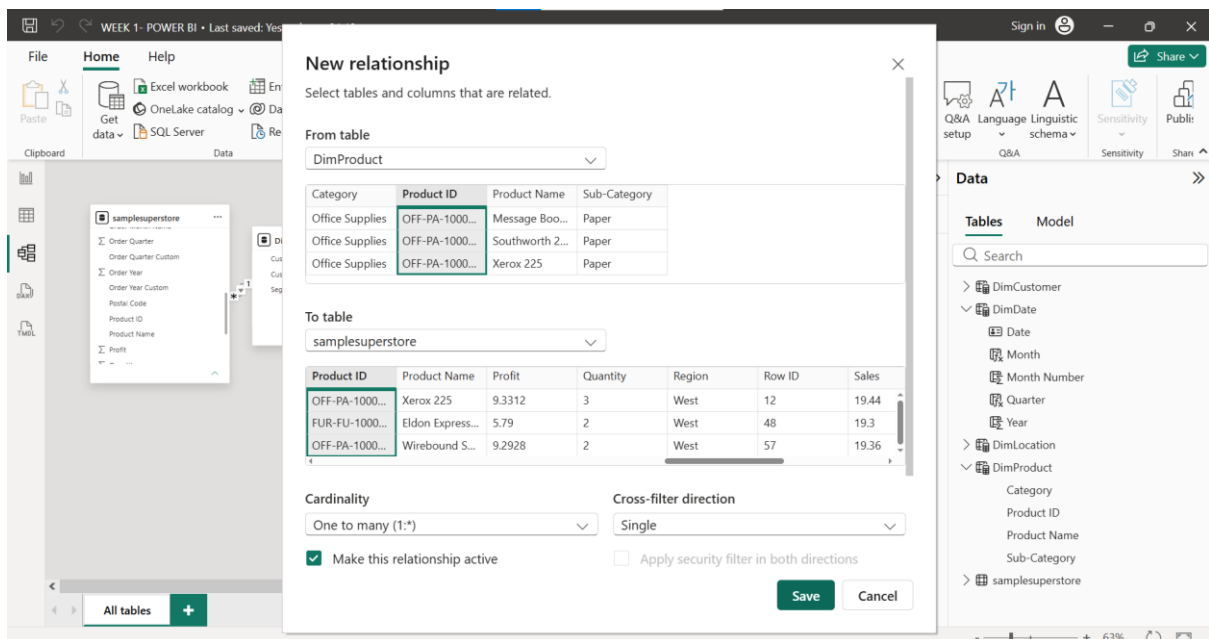
Created a **one-to-many relationship** between DimCustomer and samplesuperstore using the **Customer ID** column. This relationship connects customer information with the corresponding sales records in the fact table.



Established a one-to-many relationship between DimCustomer and the sales table using Customer ID.

12. Product Relationship

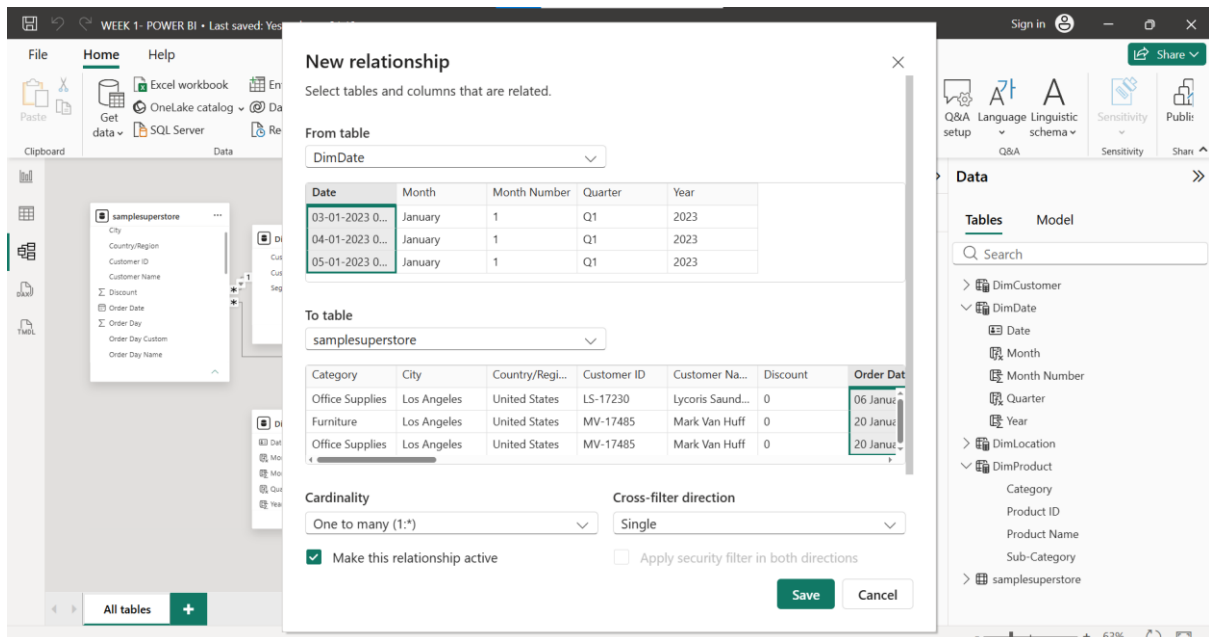
Created a **one-to-many relationship** between DimProduct and samplesuperstore using the **Product ID** column. This relationship connects product details from the dimension table with the corresponding sales records in the sales table.



Established a one-to-many relationship between DimProduct and the sales table using Product ID.

13. Date Relationship

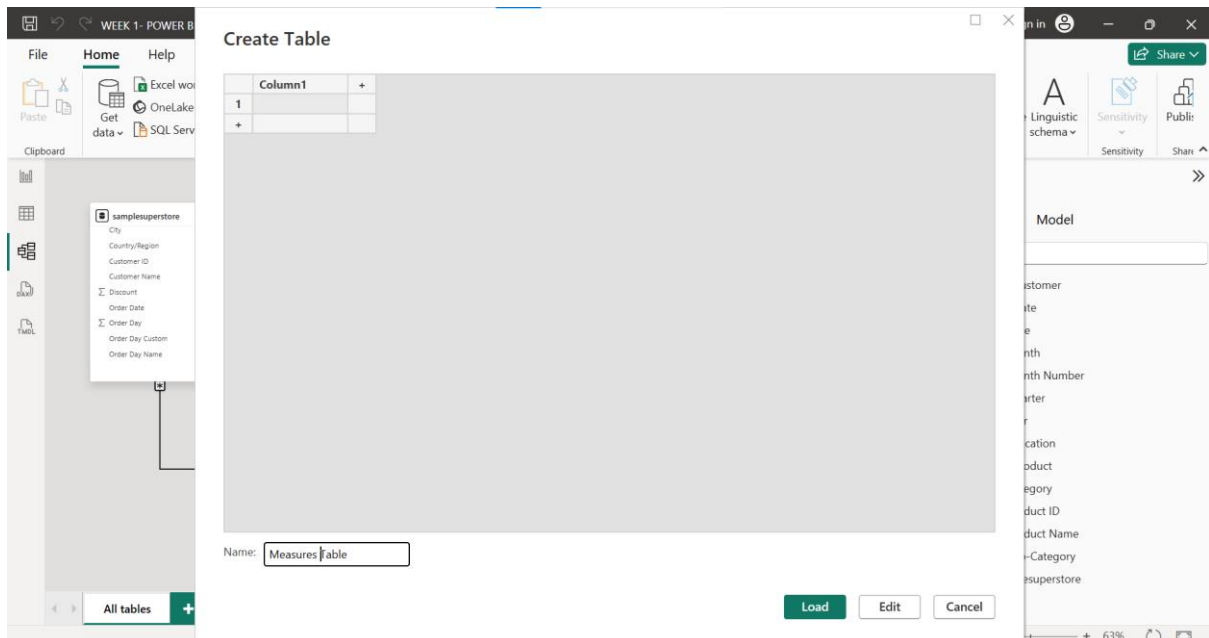
Created a **one-to-many relationship** between DimDate and samplesuperstore using the **Date** and **Order Date** columns. This relationship connects the calendar table with sales transactions and supports **time-based analysis** such as Year, Quarter, and Month.



Established a one-to-many relationship between DimDate and the sales table using Date and Order Date.

14. Measures Table Creation

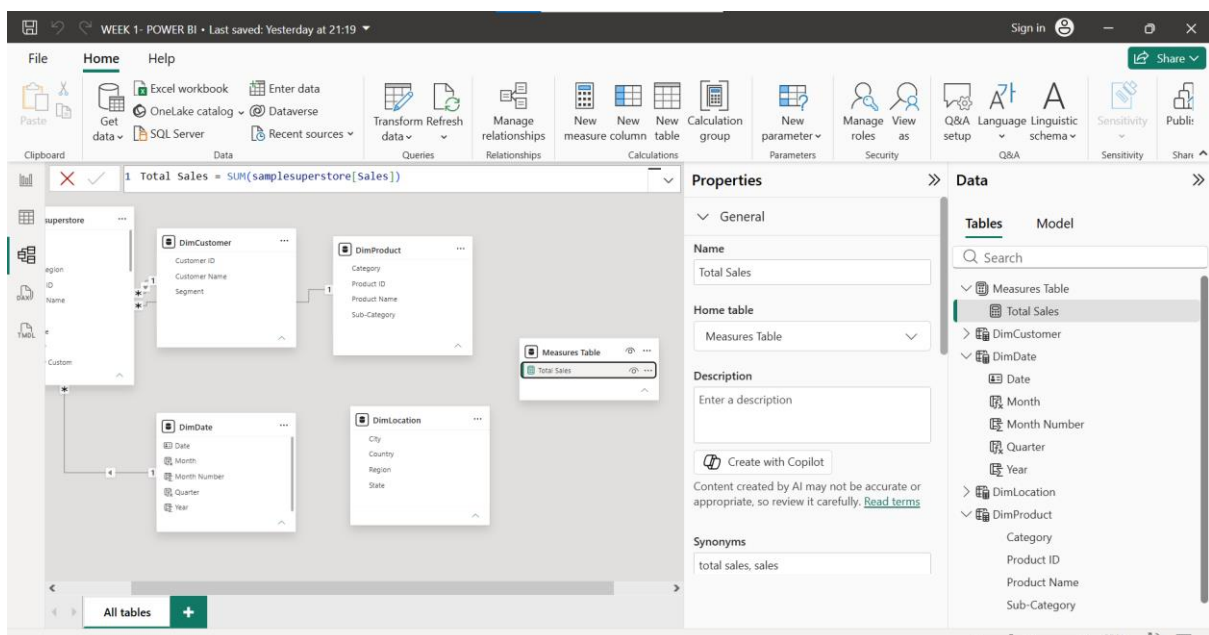
Created a dedicated **Measures table** in Power BI to store all DAX measures in one place. This helps organize calculations such as Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin.



*Created a dedicated **_Measures table** to organize and manage DAX measures.*

15. Total Sales Measure

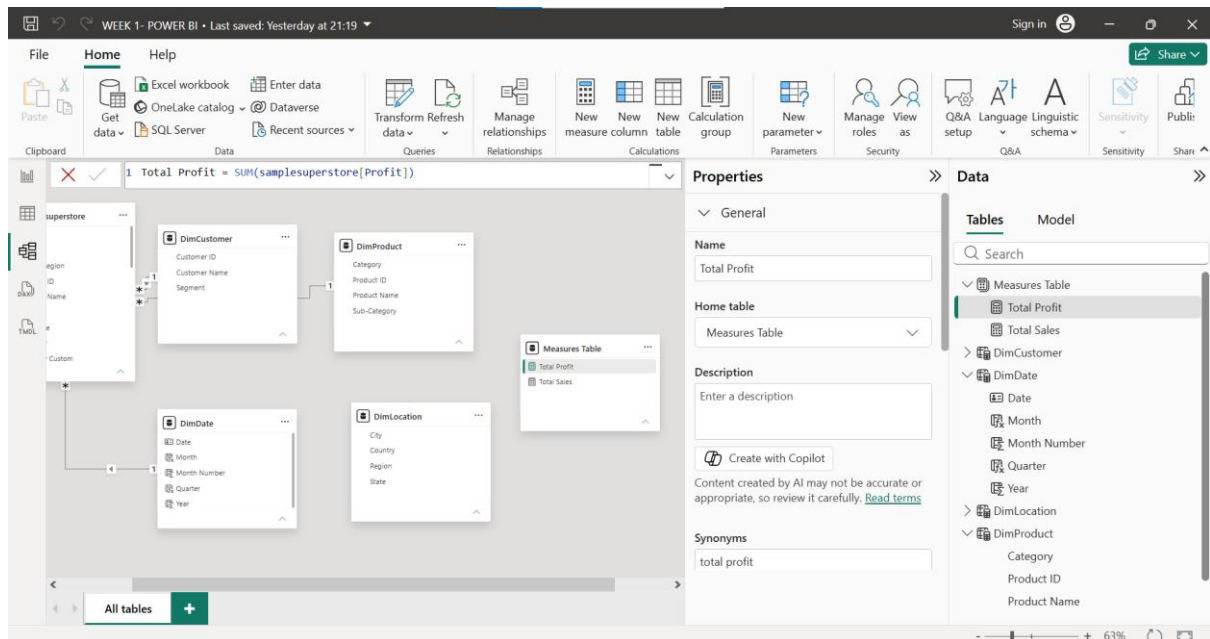
Created the **Total Sales** measure using DAX to calculate the overall sales amount from the Sales column in the samplesuperstore table.



Created the Total Sales measure to calculate overall sales performance.

16. Total Profit Measure

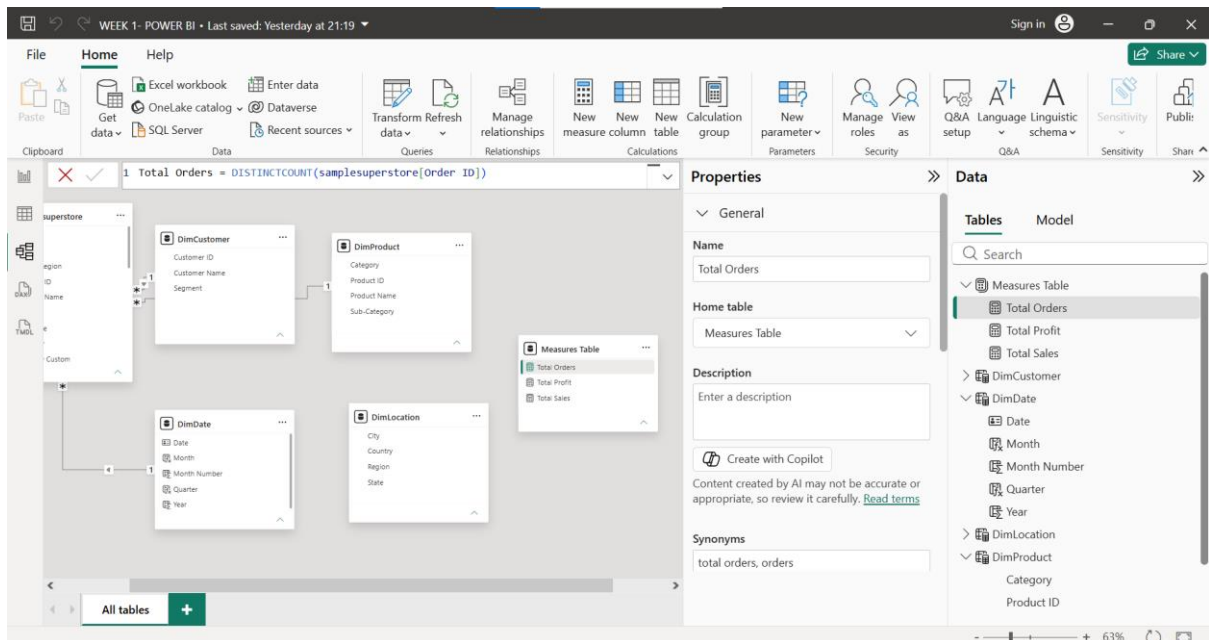
Created the **Total Profit** measure using DAX to calculate the total profit generated from all sales transactions.



Created the Total Profit measure to evaluate overall profitability.

17. Total Orders Measure

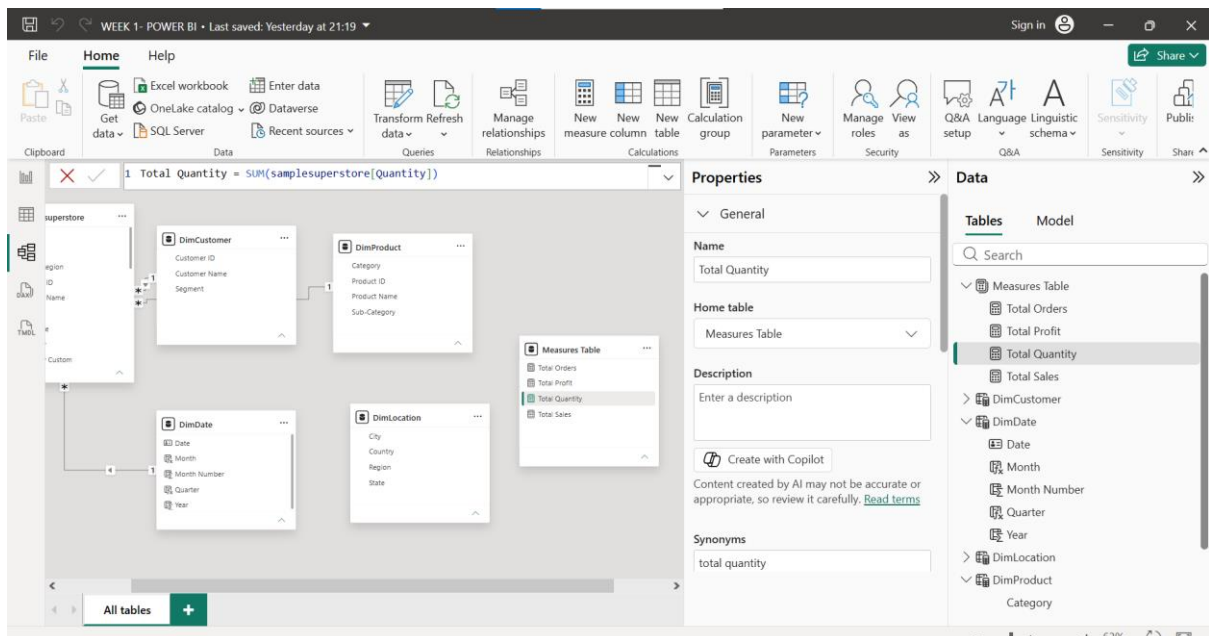
Created the **Total Orders** measure using `DISTINCTCOUNT` to calculate the number of unique orders in the `samplesuperstore` table.



Created the Total Orders measure to count unique customer orders.

18. Total Quantity Measure

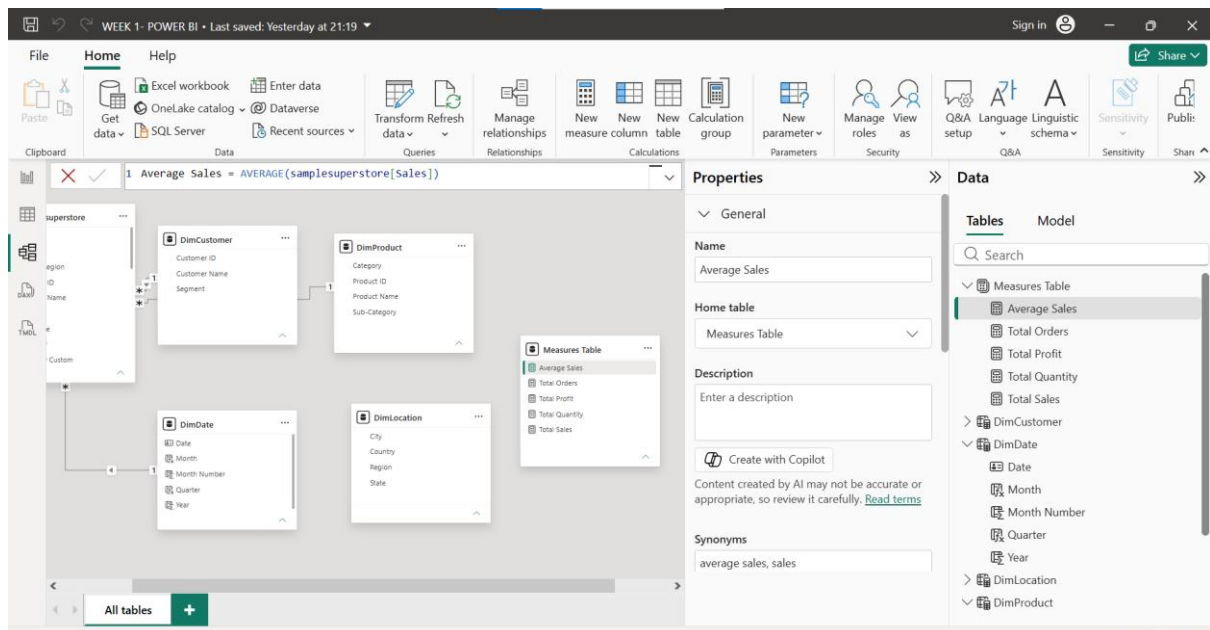
Created the **Total Quantity** measure using DAX to calculate the total number of products sold across all transactions.



Created the Total Quantity measure to analyze the overall quantity sold.

19. Average Sales Measure

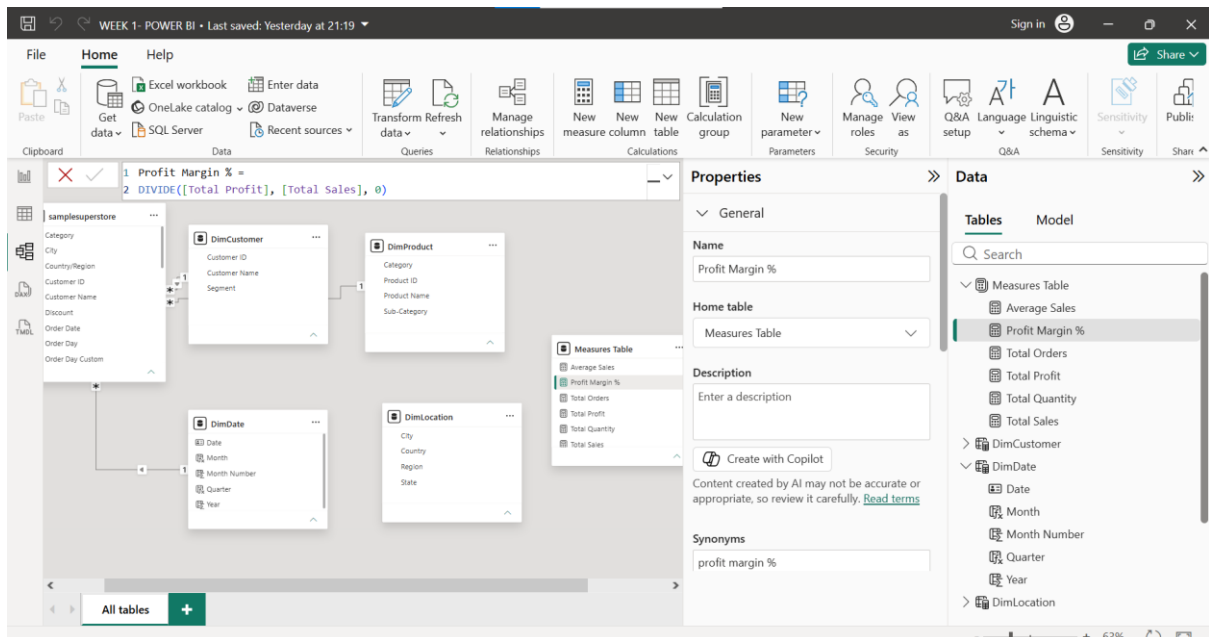
Created the **Average Sales** measure using DAX to calculate the average sales value across all sales transactions.



Created the Average Sales measure to determine the average sales value per transaction.

20. Profit Margin Measure

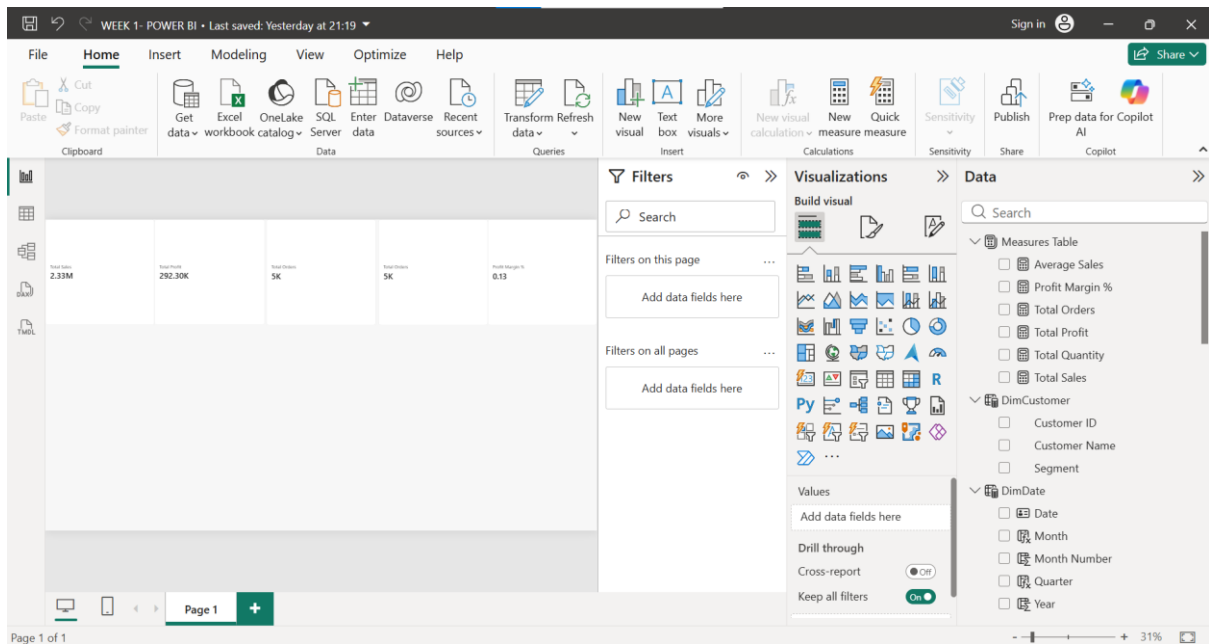
Created the **Profit Margin %** measure using the DIVIDE function to calculate profit as a percentage of total sales. The measure uses **Total Profit** divided by **Total Sales**, with 0 used when the denominator is zero.



Created the Profit Margin % measure to analyze profitability as a percentage of sales.

21. KPI Cards

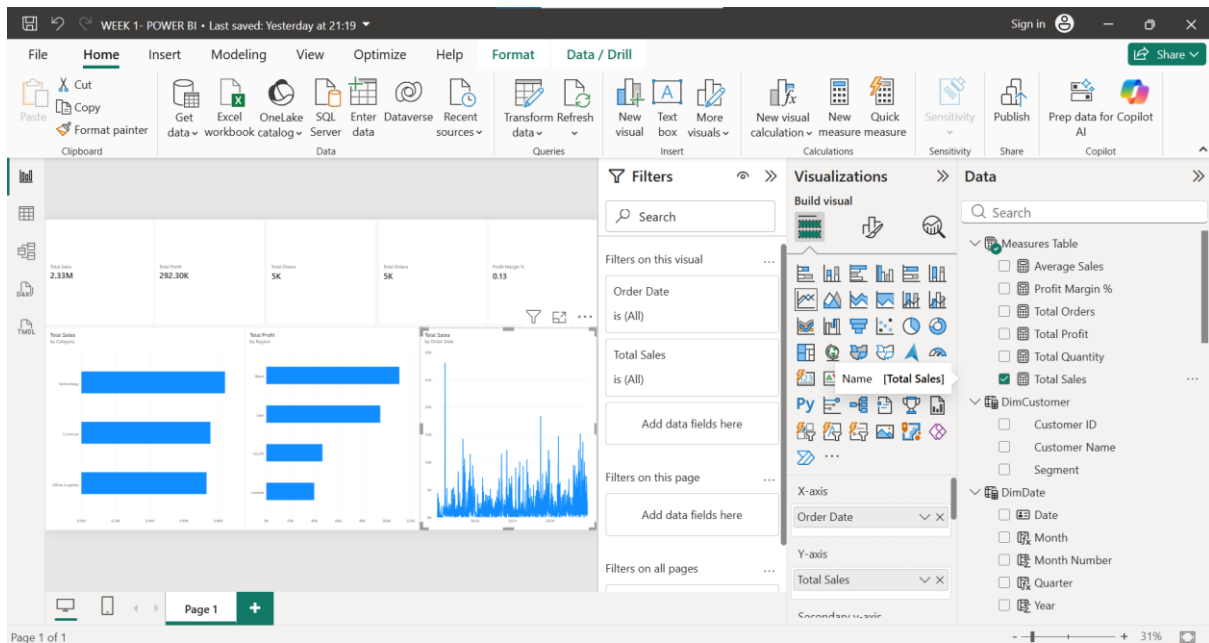
Created five KPI cards in Power BI to provide a quick overview of the key sales performance metrics. The cards display **Total Sales**, **Total Profit**, **Total Orders**, **Total Quantity**, and **Profit Margin %**, making it easier to monitor overall business performance.



Created KPI cards to display key sales, profit, order, quantity, and profit margin metrics.

22. Sales & Performance Visuals

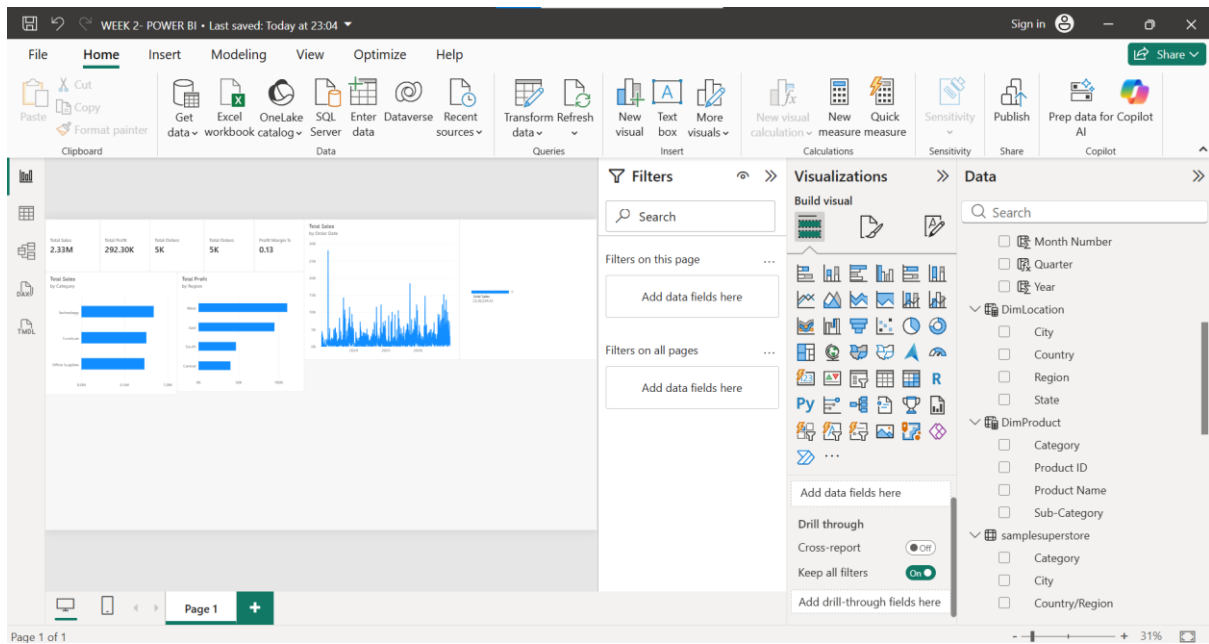
Created three visualizations in Power BI to analyze sales and profitability from different perspectives. A **Category Performance Chart** was created using Category and Total Sales, a **Regional Profitability Chart** was created using Region and Total Profit, and a **Sales Trend Line Chart** was created using Order Date and Total Sales to visualize sales performance over time.



Created category performance, regional profitability, and sales trend visuals for comprehensive sales analysis.

23. Decomposition Tree

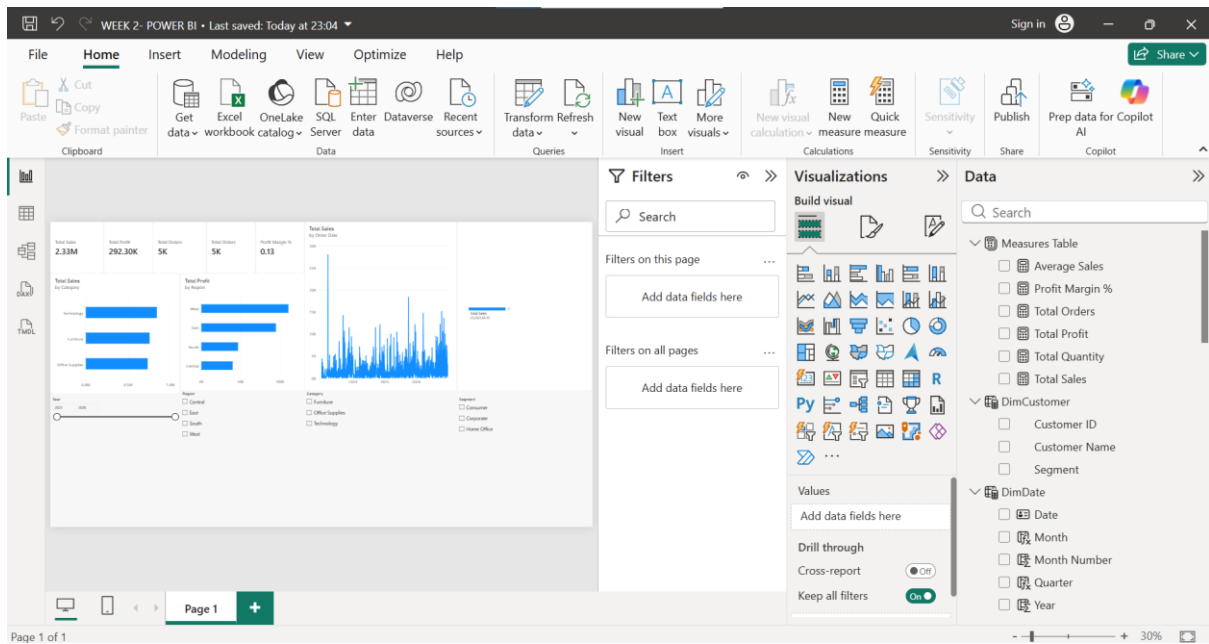
Created a **Decomposition Tree** in Power BI with **Total Sales** as the main measure and **Category, Region, Segment, and Sub-Category** as the dimensions for analysis. This interactive visual allows users to expand different categories and explore their sales performance across regions, segments, and sub-categories.



Created an interactive Decomposition Tree to analyze Total Sales by Category, Region, Segment, and Sub-Category.

24. Slicers

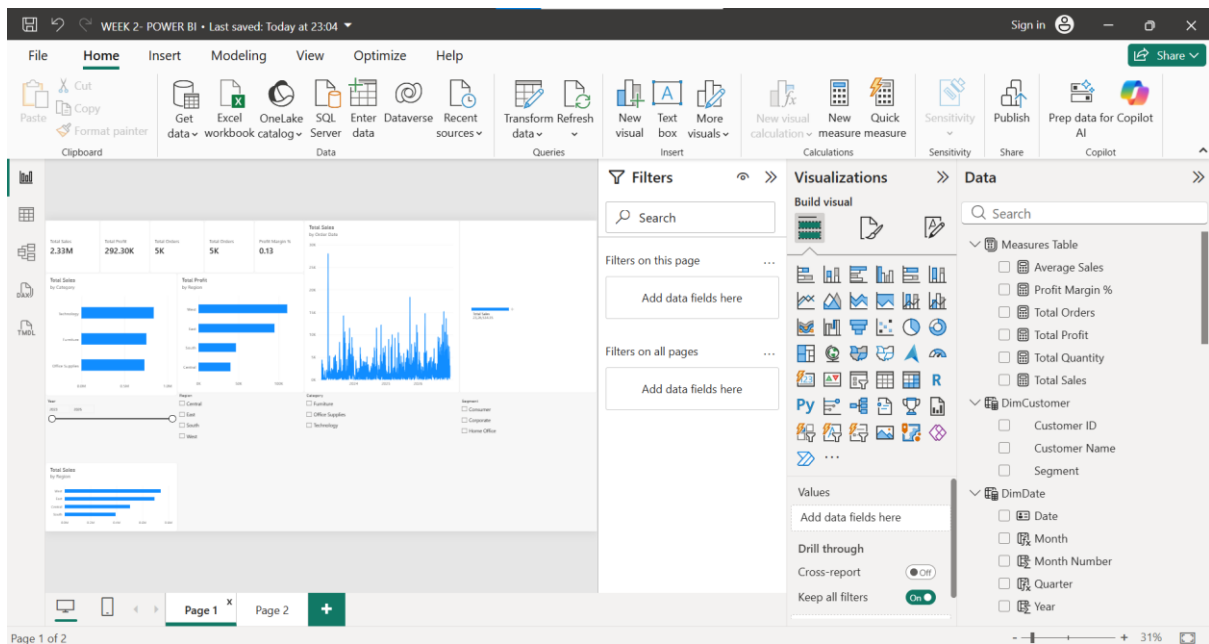
Added **four interactive slicers** to the Power BI dashboard using **Year, Region, Category, and Segment**. These slicers allow users to filter the dashboard based on their selected criteria, and the related charts and KPI cards update dynamically.



Added interactive slicers for Year, Region, Category, and Segment to enable dynamic dashboard filtering.

25. Total Sales by Region

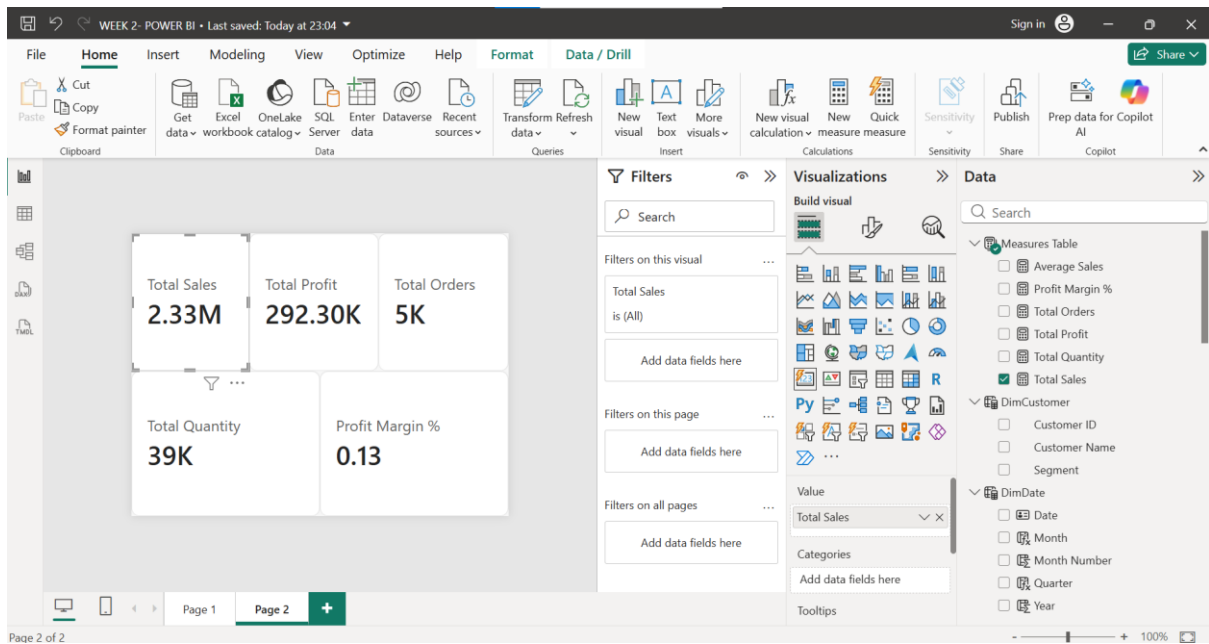
Created a **Total Sales by Region** chart to compare the sales generated across different regions. This visual helps identify regional differences in overall sales performance.



Created a regional sales performance chart using Total Sales by Region.

26. Cross-Filtering

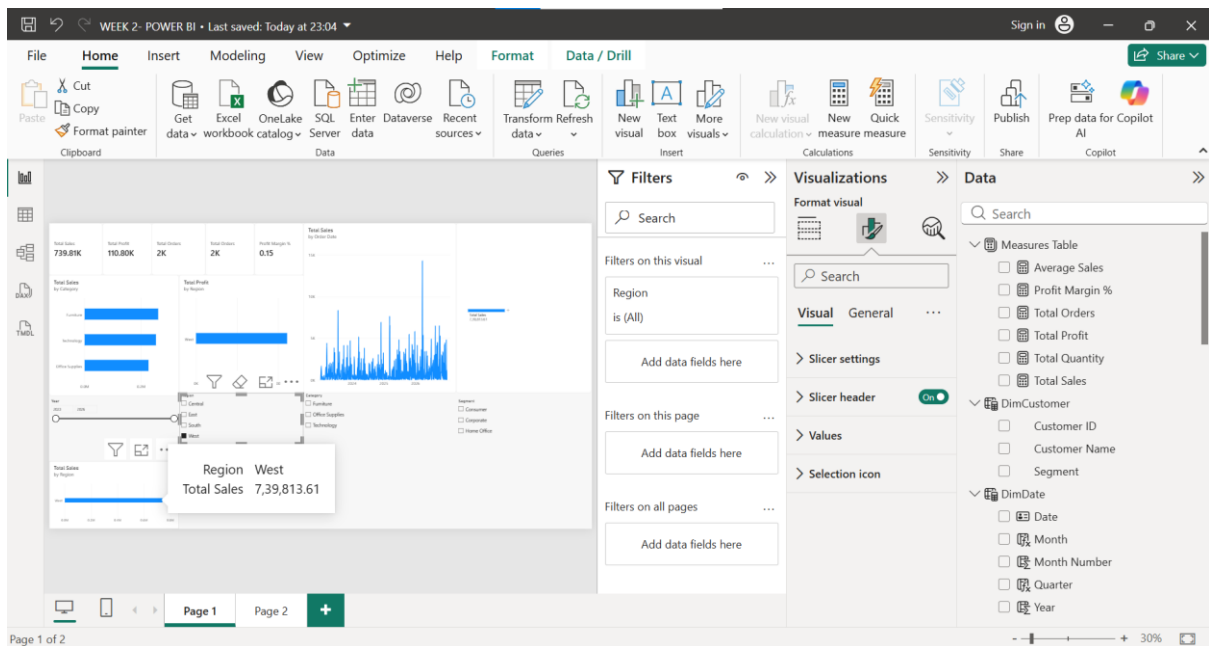
Configured and checked **cross-filtering interactions** between the visuals in the Power BI dashboard. Using **Edit interactions**, each visual was set to respond appropriately when a category or other data point is selected, allowing KPI cards, regional charts, sales trends, and other visuals to update dynamically.



Configured cross-filtering to enable interactive updates across dashboard visuals.

27. Interactive Tooltip

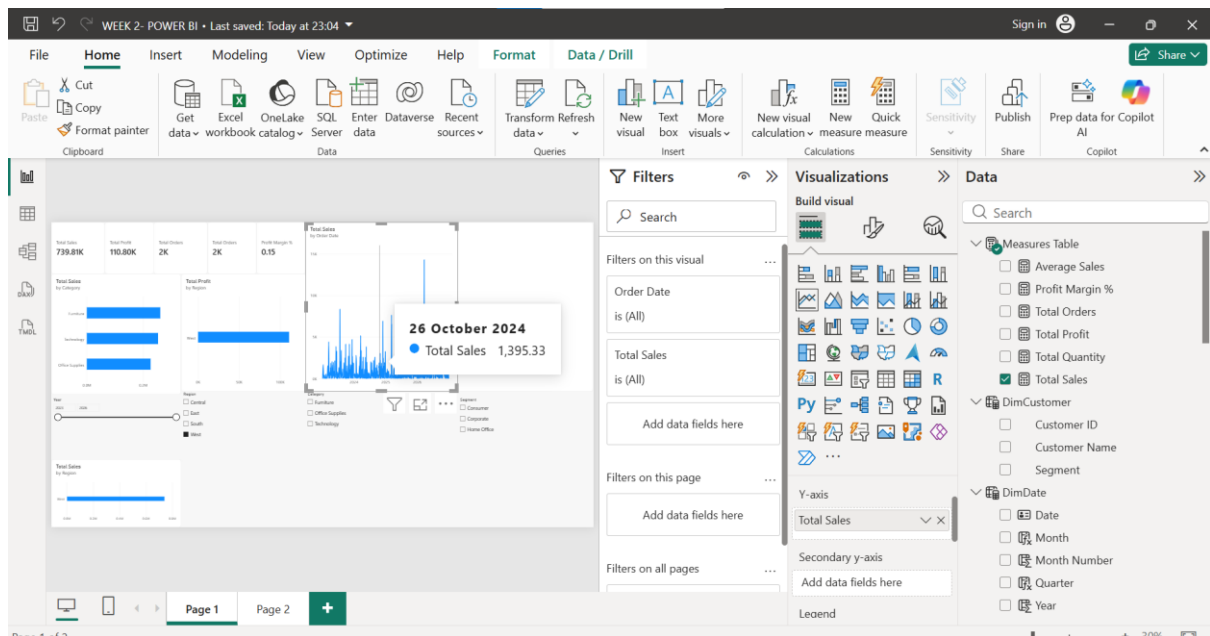
Created a **Report Page Tooltip** in Power BI to display additional information when the user hovers over a data point. The tooltip provides useful details such as **Region**, **Total Sales**, **Total Profit**, **Profit Margin %**, and **Total Orders**, making the dashboard more interactive and informative.



Created an interactive report page tooltip to display detailed sales and profitability information on hover.

28. Drill-Down

Created a **hierarchical drill-down structure** in the Power BI dashboard to enable detailed analysis at multiple levels. The hierarchy can be explored from **Year → Quarter → Month** or **Category → Sub-Category → Product**, allowing users to move from a summary view to more detailed information interactively.



Created drill-down hierarchies for detailed time-based and category-level analysis.